# Synopsis: CollabStack IDE – Real-Time Collaborative Development Platform

## Introduction:

CollabStack IDE is an advanced real-time collaborative development platform that enables multiple developers to code together, execute programs on distributed compute resources, and leverage integrated AI assistance within a unified workspace. It is designed to eliminate environment setup friction and make team-based software development faster, more interactive, and infrastructure-free.

## Problem Statement:

Traditional development workflows rely on local IDEs, manual environment setup, and asynchronous collaboration through version control, which often leads to configuration issues, slow feedback cycles, and communication gaps between team members. Existing online editors typically focus only on code editing and lack deep integration with execution environments, design tools, and AI assistance. This fragmented ecosystem makes real-time collaboration, distributed execution, and end-to-end project coordination difficult for modern software teams.

## Solution Overview:

CollabStack IDE addresses these challenges by combining real-time code collaboration, distributed code execution, AI-powered coding assistance, and productivity integrations into a single browser-based platform. Users can jointly edit files with live cursor tracking, run code on remote CPU/GPU clusters, visualize resource usage, and interact with an AI assistant that understands project context. The system integrates with GitHub, Google Drive, Google Calendar/Meet, and email, offering a seamless workflow from code editing to communication and deployment.

## Key Features:

- **Real-time collaboration:** Multi-user live code editing with cursor tracking, presence indicators, and synchronized file states.
- **Distributed execution:** Remote execution on CPU/GPU clusters with real-time output streaming and resource metrics.
- **AI assistance:** Context-aware AI chat for code explanation, debugging help, and code suggestions based on project files.
- **Integrated design canvas:** Figma-like collaborative design board stored alongside code for design–development alignment.

● **Multi-language support:** Execution support for 17+ languages with terminal tabs and built-in package management commands.
● **Seamless integrations:** Auto-push to GitHub, Google Drive sync, Google Meet link generation, and calendar event creation.
● **Security & auth:** OTP-based email verification, JWT authentication, secure password hashing, and OAuth2-based integrations.

## Technology Stack:

● **Frontend**: React, TypeScript, Monaco Editor, Socket.IO, Tailwind CSS, Zustand, Framer Motion.
● **Backend:** Node.js, Express.js, Socket.IO (real-time), Prisma ORM, PostgreSQL database.
● **External Services:** Judge0 API for code execution, OpenAI API for AI assistance, Google APIs (Drive, Calendar, Meet), SMTP for email/OTP.
● Tools & Infrastructure: RESTful APIs, WebSockets for real-time communication, JSON-based data exchange, role- and project-based data models.

## Future Scope:

● Addition of video calling and screen sharing for richer in-IDE communication.
● Plugin and extension ecosystem for custom tools, linters, and domain-specific workflows.
● Enhanced AI capabilities such as automatic refactoring, test generation, and multi-file code generation.
● Offline and low-connectivity modes with conflict resolution and background synchronization.
● Deeper integration with CI/CD pipelines, container orchestration, and monitoring tools for complete DevOps workflows.

## Conclusion:

CollabStack IDE modernizes collaborative software development by unifying real-time coding, distributed execution, AI assistance, and cloud integrations into a single platform. By reducing setup overhead, improving visibility, and enabling truly synchronous collaboration, it serves as a robust and industry-relevant solution suitable for academic projects and real-world team development environments.